NINE EDUCATION / CLASS XI PHYSICS

THERMODYNAMICS

40 JEE Main Numericals

Subjective questions with precision P–V diagrams

THE PRACTICE WORKBOOK

First Law • Molar Heat Capacities • γ • Gas Mixtures

40	20+20	0
Numerical problems	**Two lecture sets**	**Multiple-choice options**

All diagrams are drawn as vector graphics and remain sharp at any zoom level.

Student name: _____ Section: _____

Suggested duration: 2 hours 30 minutes

Answer key at the end

BEFORE YOU BEGIN

Instructions and conventions

- All questions require **numerical answers with working**. No options are provided.
- All gases are ideal, and all paths are quasistatic, unless stated otherwise.
- Work done **by the gas** is positive; for the first law, use $Q = \Delta U + W$.
- Use $R = 8.3\ \text{J mol}^{-1}\text{K}^{-1}$, $\ln 2 = 0.693$, $\ln 4 = 1.386$, $\ln 8 = 2.079$.
- $1\ \text{L} = 10^{-3}\ \text{m}^3$. Temperature differences in kelvin and Celsius have equal numerical values.
- Unless otherwise noted, monatomic $f = 3$, diatomic $f = 5$, and non-linear polyatomic $f = 6$; vibrational modes are inactive.
- For diagrams, P and V are plotted to scale; arrowheads show the direction of the process. When only end states are shown, the path is deliberately unspecified.

Formula reference

$$\Delta U = nC_V\Delta T = \frac{f}{2}(P_fV_f - P_iV_i), \qquad Q = \Delta U + \int P\,dV$$

$$C_V = \frac{f}{2}R, \qquad C_P = C_V + R, \qquad \gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f}$$

$$C_{V,\text{mix}} = \frac{\sum_i n_i C_{V,i}}{\sum_i n_i}, \qquad \gamma_{\text{mix}} = 1 + \frac{R}{C_{V,\text{mix}}}$$

Two-part coverage

01–20 Energy balance, P–V work, isobaric / isochoric / isothermal paths, cycles.

21–40 C_V, C_P, Mayer's relation, γ, mole-weighted mixtures, and combined P–V problems.

Teacher's note. Questions with graphical information use bespoke vector plots rather than raster screenshots. The answer key includes results and compact verification steps, and begins on its own pages.

PART 01

First Law and Processes

Questions 01–20 / Energy, work, internal energy, and P–V graphs

QUESTION 01

A rigid vessel contains 2 moles of a monatomic ideal gas at $300\,\text{K}$. The gas is heated to $480\,\text{K}$. Calculate the **heat supplied** and the **work done by the gas**.

QUESTION 02

A rigid container holds 3 moles of a diatomic ideal gas at $400\,\text{K}$. Its temperature is increased to $500\,\text{K}$. Find the **heat absorbed** and the **ratio of final to initial pressure**.

QUESTION 03

A gas absorbs $1800\,\text{J}$ of heat and performs $650\,\text{J}$ of work. Calculate its **change in internal energy**. If the initial internal energy was $2400\,\text{J}$, determine its final value.

QUESTION 04

An external agent does $2500\,\text{J}$ of work **on** a gas during compression. Simultaneously, the gas releases $900\,\text{J}$ of heat. Calculate the change in internal energy of the gas, with sign.

QUESTION 05

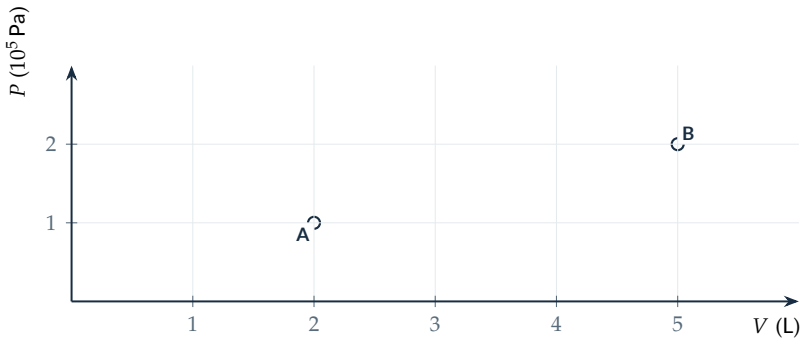
Two moles of a monatomic ideal gas are heated from $300\,\text{K}$ to $450\,\text{K}$ at constant pressure. Calculate the **work done**, **change in internal energy**, and **heat supplied**.

QUESTION 06

A monatomic ideal gas requires $7470\,\text{J}$ of heat to undergo a certain temperature rise at constant pressure. Find the heat required for the **same temperature rise at constant volume** and the difference between the two heat inputs.

QUESTION 07

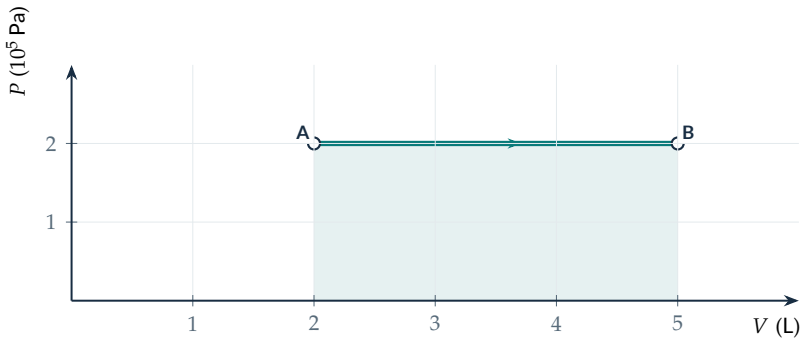
A monatomic ideal gas changes from state A to state B, as shown. Determine the change in internal energy, **regardless of the path** followed between these states.



Only the initial and final states are shown; no process path is specified.

QUESTION 08

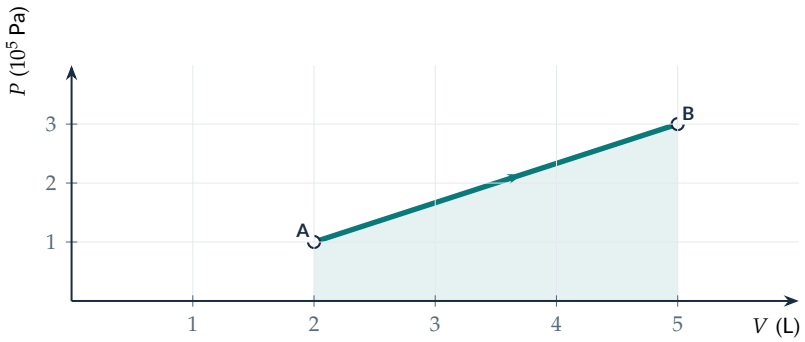
A monatomic ideal gas expands at constant pressure along the indicated path from A to B. Determine the **work done**, **change in internal energy**, and **heat absorbed**.



Horizontal P–V path: constant pressure at 2×10^5 Pa.

QUESTION 09

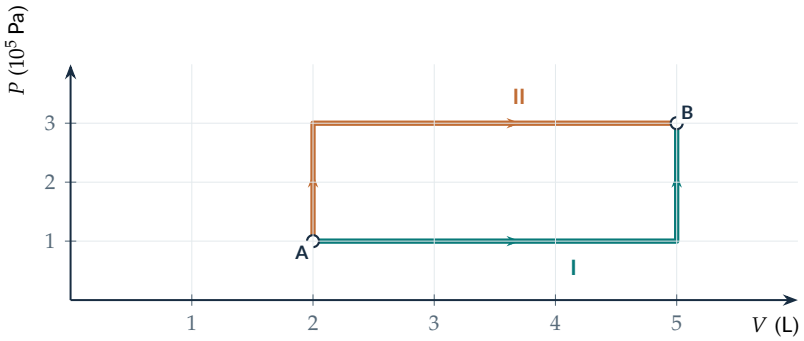
A monatomic ideal gas expands quasistatically along the straight path $A \rightarrow B$. Determine the **work done by the gas** and **total heat absorbed**.



Pressure varies linearly between the labeled endpoints.

QUESTION 10

A monatomic ideal gas goes from A to B by two different quasistatic paths shown below. Path I expands at low pressure before heating at fixed volume; Path II heats at fixed volume first and then expands. Calculate the **work and heat** along each path, and the difference $Q_{\text{II}} - Q_{\text{I}}$.



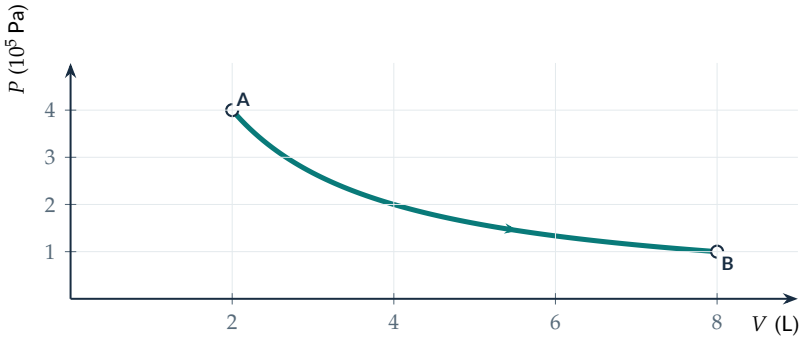
Teal: Path I. Orange: Path II. Both paths have the same endpoints.

QUESTION 11

An ideal gas expands isothermally from V to $2V$. Its initial pressure-volume product is $PV = 2000\,\text{J}$. Calculate the **work done by the gas** and the **heat absorbed**.

QUESTION 12

An ideal gas initially occupies $2\,\text{L}$ at 4×10^5 Pa. It slowly expands isothermally along the curve shown until its volume becomes $8\,\text{L}$. Find the **final pressure** and the **work done by the gas**.



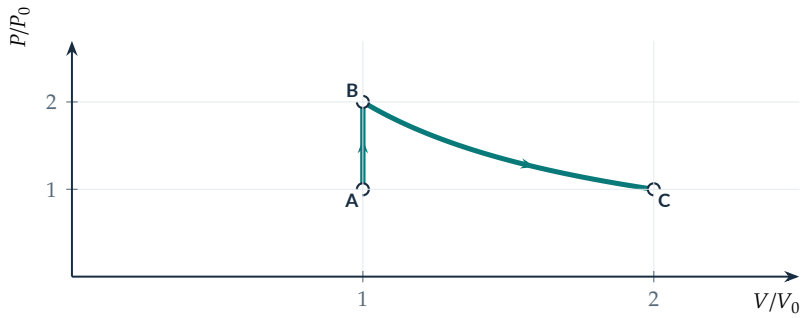
Isotherm $PV = 8 \times 10^5\ \text{Pa} \cdot \text{L}$.

QUESTION 13

An ideal gas is compressed isothermally from V to $V/2$. If initially $PV = 1200\,\text{J}$, find the **work done on the gas** and the **heat released to the surroundings**.

QUESTION 14

One mole of monatomic ideal gas initially at $300\,\text{K}$ is heated at constant volume to $600\,\text{K}$ and then expands isothermally until its volume doubles. Calculate the **total work done** and the **total heat absorbed**.



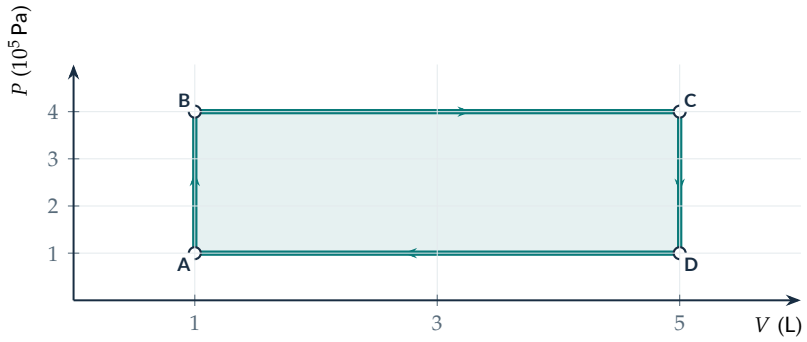
Schematic normalized graph; $A \to B$ is isochoric and $B \to C$ is isothermal.

QUESTION 15

Two moles of monatomic ideal gas at $400\,\text{K}$ are heated at constant pressure to $600\,\text{K}$, then cooled at constant volume to $500\,\text{K}$. Find the **net work**, **net change in internal energy**, and **net heat absorbed**.

QUESTION 16

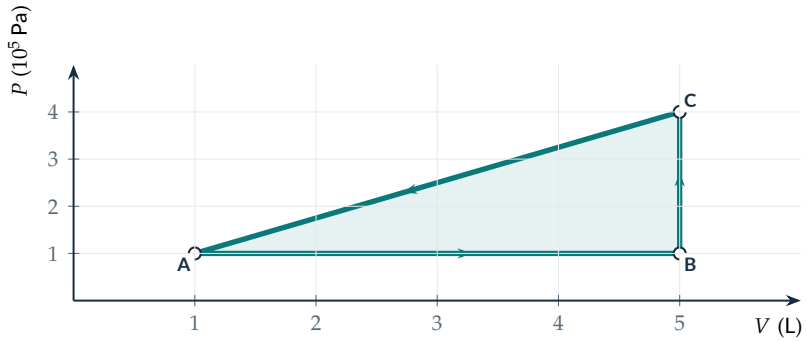
An ideal gas executes the **clockwise rectangular cycle** illustrated below. Determine the **net work done by the gas** and the **net heat absorbed** in one complete cycle.



Clockwise cycle; the enclosed area is the net positive work.

QUESTION 17

An ideal gas goes through $A \to B \to C \to A$ on the graph. The segment $C \to A$ is a straight line. Calculate **net work done by the gas** and **net heat absorbed** over the cycle.



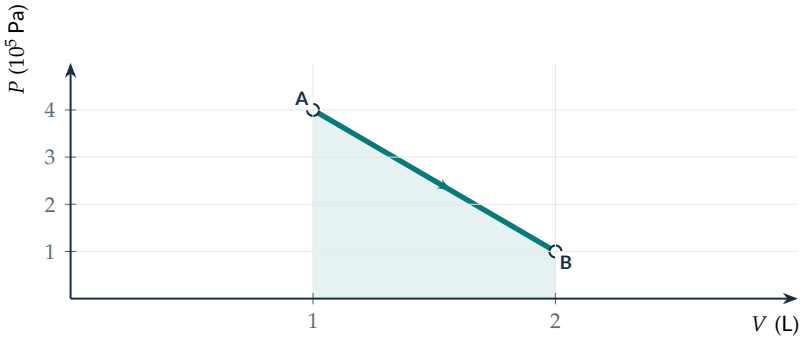
Follow the arrows: this triangular cycle is anticlockwise.

QUESTION 18

An ideal gas initially occupies $8\,\text{L}$ at a pressure of 1×10^5 Pa. It is slowly compressed isothermally until the pressure becomes eight times its initial value. Determine **final volume**, **work done by the gas**, and **heat exchanged** (including sign).

QUESTION 19

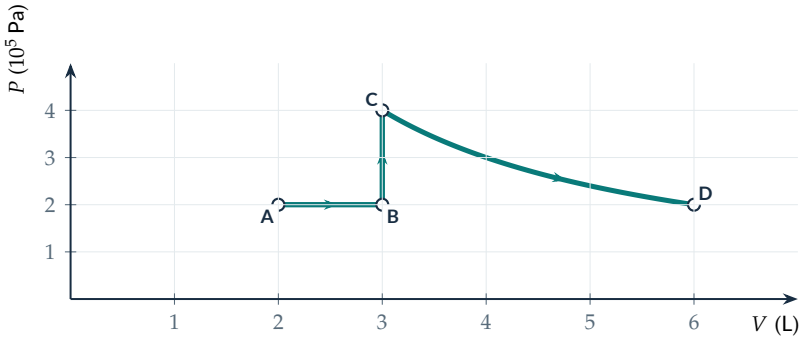
A monatomic ideal gas expands along the straight line $A \to B$ shown below. Determine **work done**, **change in internal energy**, and **heat exchanged**. State whether heat is absorbed or released although the gas expands.



A falling-pressure straight-line expansion: expansion does not guarantee heat absorption.

QUESTION 20

A monatomic ideal gas starts at A and follows $A \to B$ at constant pressure, $B \to C$ at constant volume, and then $C \to D$ isothermally until its pressure returns to 2×10^5 Pa. Find the **total work**, **total change in internal energy**, and **net heat absorbed**.



The final curved leg is the isotherm $PV = 1200\,\text{J}$.

PART 02

Heat Capacity and Mixtures

Questions 21–40 / C_V, C_P, γ, Mayer's relation, and mole-weighted averages

QUESTION 21

Two moles of diatomic ideal gas are heated from $293\,\text{K}$ to $353\,\text{K}$. Determine C_V and C_P in $\text{J mol}^{-1}\text{K}^{-1}$. Then calculate the heat needed at **constant volume**, heat needed at **constant pressure**, and their difference.

QUESTION 22

A monatomic ideal gas containing 3 moles expands at constant pressure and does $1494\,\text{J}$ of work. Calculate the **temperature rise**, **heat supplied**, and **increase in internal energy**.

QUESTION 23

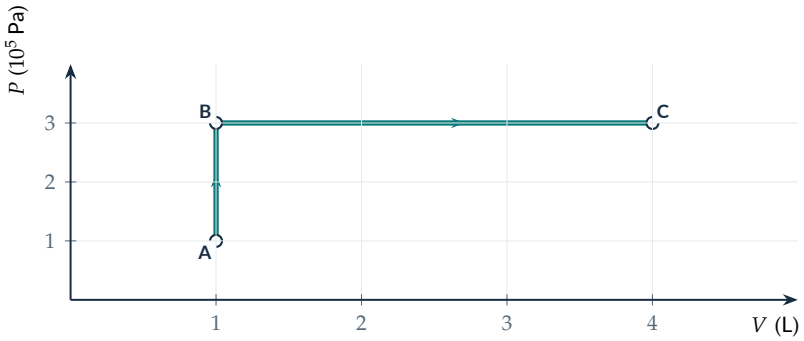
Two moles of an unknown ideal gas absorb $3320\,\text{J}$ of heat at constant pressure while their temperature rises from $200\,\text{K}$ to $280\,\text{K}$. Find its **molar heat capacity** C_P, **degrees of freedom** f, and **heat-capacity ratio** γ.

QUESTION 24

A gas has $\gamma = 4/3$ and absorbs $1600\,\text{J}$ at constant pressure. Calculate the **increase in internal energy**, the **work done by the gas**, and its effective number of degrees of freedom.

QUESTION 25

A diatomic ideal gas passes through states $A \to B \to C$ on the indicated P–V graph. Calculate (i) the **heat supplied in** $A \to B$, (ii) the **heat supplied in** $B \to C$, and (iii) the **total heat supplied**.



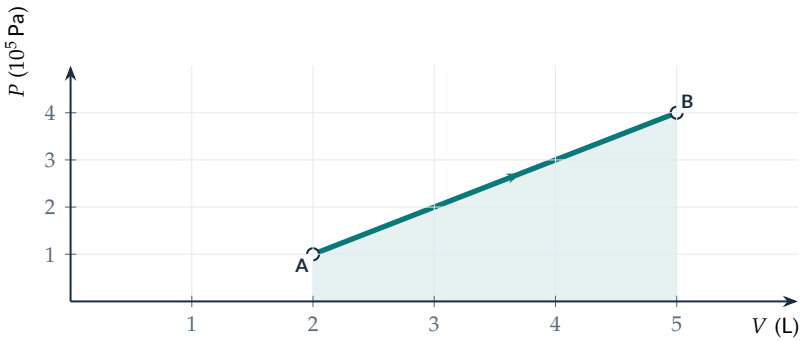
Heating at fixed volume is followed by expansion at fixed pressure.

QUESTION 26

One mole of oxygen (O_2) receives $2075\,\text{J}$ of heat while held at constant volume. Calculate its **temperature rise**. How much heat would it require for the **same temperature rise at constant pressure**?

QUESTION 27

A diatomic ideal gas moves quasistatically from A to B on the straight-line path shown. Find the **work done**, **change in internal energy**, and **total heat absorbed**.



Use the trapezoidal area for work, but C_V for ΔU even though V changes.

QUESTION 28

A mixture contains 2 moles of helium (monatomic) and 3 moles of hydrogen (diatomic). Assuming ideal-gas behavior, determine its **effective degrees of freedom**, **molar** C_V, **molar** C_P, and γ_{mix}.

QUESTION 29

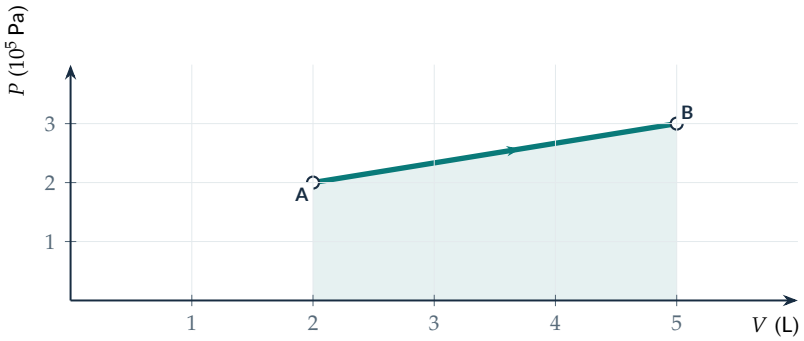
Three moles of a monatomic ideal gas are mixed with n moles of a diatomic ideal gas. If the mixture has $\gamma = 29/19$, find n and the **effective degrees of freedom** of the mixture.

QUESTION 30

A rigid vessel contains a mixture of 1 mole of helium and 2 moles of nitrogen, initially at 300 K. The mixture is heated to 420 K. Calculate (i) the **heat absorbed**, (ii) the **final-to-initial pressure ratio**, and (iii) γ_{mix}.

QUESTION 31

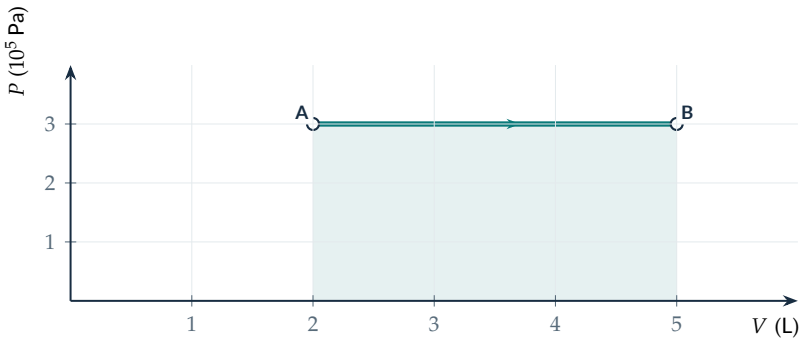
The same P–V path $A \to B$, shown below, is followed separately by two gases: Gas X is monatomic; Gas Y is diatomic. For each gas, calculate the **work done** and the **heat absorbed**. Find $Q_Y - Q_X$.



Identical end states and the same path, but the gas degrees of freedom differ.

QUESTION 32

An unknown ideal gas expands at constant pressure along the line shown. The **heat absorbed** during the process is $2250\,\text{J}$. Calculate its **work done**, **change in internal energy**, **degrees of freedom**, and γ.



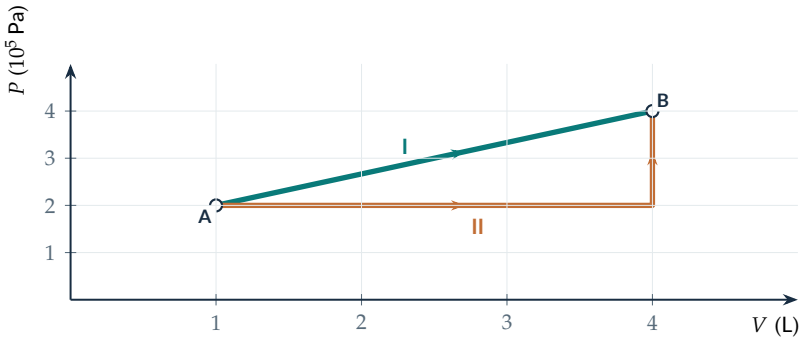
Use the P–V data and the first law to infer the type of gas.

QUESTION 33

A mixture of 2 moles of helium and 2 moles of oxygen is heated through $50\,\text{K}$ at constant pressure. Calculate the **heat supplied**, **work done**, **rise in internal energy**, and γ_{mix}.

QUESTION 34

A diatomic ideal gas goes from A to B along the two paths shown: Path I is a straight line; Path II expands at low pressure and then increases pressure at constant volume. Calculate **work done** and **heat absorbed** along each path. Find $Q_{\text{I}} - Q_{\text{II}}$.



Compare the areas under the paths; ΔU is the same.

QUESTION 35

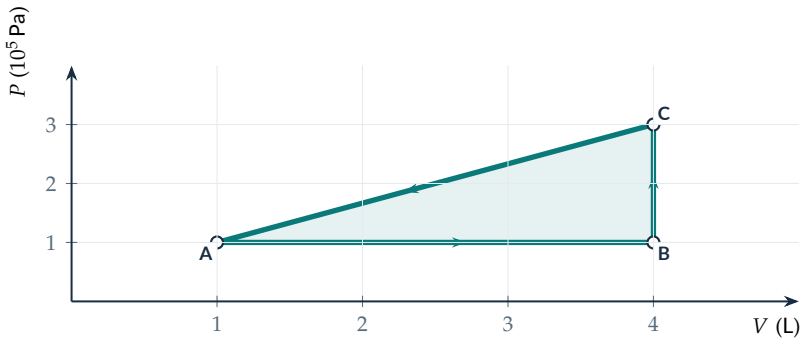
A thermally insulated, rigid vessel has two compartments separated by a removable insulated partition. One compartment holds 1 mole of helium at $600\,\text{K}$, and the other contains 2 moles of nitrogen at $340\,\text{K}$. The partition is removed and the gases mix without heat exchange with the surroundings. Determine their **final equilibrium temperature**.

QUESTION 36

Two moles of a monatomic ideal gas are mixed with n moles of a non-linear polyatomic ideal gas (six active degrees of freedom). The mixture has $\gamma = 7/5$. Find n and the **molar heat capacities** C_V and C_P of the mixture.

QUESTION 37

A diatomic ideal gas is taken once around the triangular cycle $A \to B \to C \to A$ shown. Calculate the **heat absorbed in each leg** and the **net heat absorbed** over the cycle.



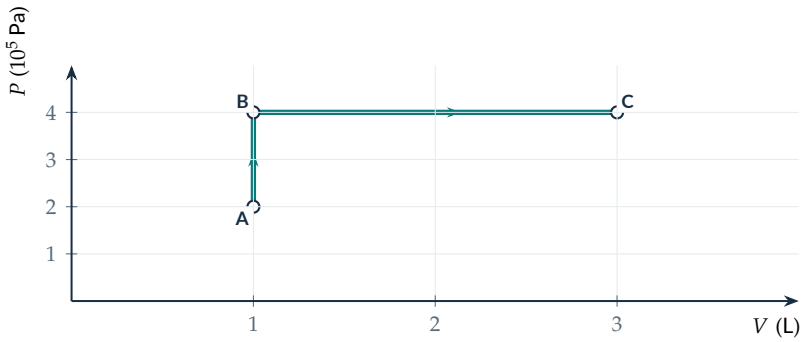
The return leg $C \to A$ is a straight line (not an isotherm).

QUESTION 38

One mole of a gas with $\gamma = 5/3$ is mixed with two moles of another gas with $\gamma = 7/5$. Calculate $C_{V,\text{mix}}$, $C_{P,\text{mix}}$, and γ_{mix}. Hence determine the **heat required** to raise the mixture temperature by $100\ \text{K}$ at constant pressure.

QUESTION 39

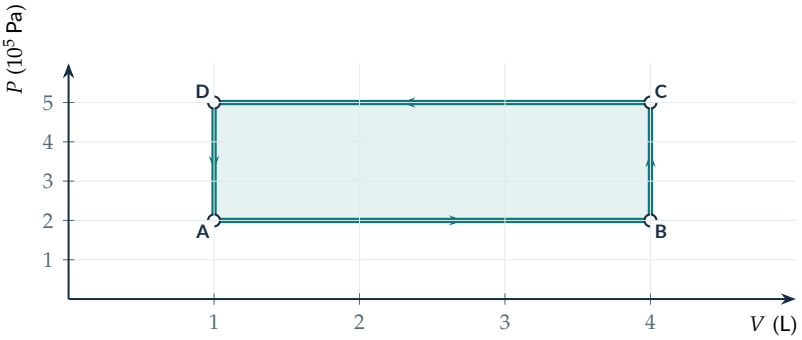
A mixture of 3 moles of monatomic gas and 1 mole of diatomic gas is taken through the two-step path $A \to B \to C$ illustrated. Determine (i) γ_{mix}, (ii) **heat absorbed in each leg**, and (iii) **total heat absorbed**.



Mole-weighted internal energy first; then apply the first law to each leg.

QUESTION 40

A mixture of 3 moles of a monatomic ideal gas and 2 moles of a diatomic ideal gas executes the four-step cycle $A \to B \to C \to D \to A$. Determine (i) γ_{mix}, (ii) **heat exchanged on each leg** with signs, and (iii) the **net work and net heat** for the complete cycle.



Anticlockwise rectangular cycle; γ_{mix} must be found by mole-weighting C_V.

WORKING SPACE FOR QUESTION 40